Programme Budgeting and Marginal Analysis (PBMA)

1. Prioritisation

As a tax-funded system the NHS is under constant strain, as the demand for healthcare is close to infinite, whilst healthcare resources are finite.

To create a balance between finite resources and infinite wants and needs, choices are necessary and consequently costs and benefits must be compared.

Prioritisation is also required to assist this decision making so resources are used in a way that maximises their benefit.

2. Programme Budgeting and Marginal Analysis (PBMA)

Programme Budgeting and Marginal Analysis (PBMA) is a resource allocation framework which can be applied at a macro and micro level to help decision-makers allocate resources so that benefits are maximised.

Programme budgeting is an appraisal of current resource allocation in specified set of programmes or activities, with a view to tracking future resource allocation in those same programmes/ activities.

Marginal analysis is the appraisal of the added benefits and added costs of a proposed investment or the lost benefits and lower costs of a proposed disinvestment.

This is often referred to by health economists as ***'opportunity costs'***, the value of benefits foregone by not using resources in their next best alternative use (i.e. the next best thing we could have done with the money).

This framework results in recommendations of where investment/ disinvestment should be made. These recommendations can then be actioned by directors and managers.

3. Stages of PBMA

Brambleby and Fordham describe eight stages to conducting a PBMA[1] [2]

1. Choose a set of meaningful programmes
2. Identify current activity and expenditure in those programmes
3. Think of improvements
4. Weigh up incremental costs and incremental benefits and prioritize a list
5. Consult widely
6. Decide on changes
7. Effect the changes

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[1] Brambleby P, Fordham R: What is PBMA? 4 (2): http://www.medicine.ox.ac.uk/bandolier/painres/download/whatis/pbma.pdf,Return to ref 8 in article

[2] Brambleby P, Fordham R: Implementing PBMA? 4 (3): http://www.medicine.ox.ac.uk/bandolier/painres/download/whatis/pbmaimp.pdf,

8. Evaluate progress

3.1 Stage 1 – Choose a set of meaningful programmes

The choice of programmes could be dictated by a particular budget or a particular initiative e.g. improving mental health in adults aged 18 years and above. A manager or director may ask for the exercise to be carried out on a specific set of programmes, activities or on a pathway. A PBMA could also be requested to assess where cost savings could be made.

It is important to establish the boundary of the PBMA exercise. If activity occurs across boundaries it can be difficult to attribute spending and outcomes to specific activities. It is best to use a budget or programme where the boundaries are clearly defined.

At this stage it is wise to establish the PBMA panel who will make investment and disinvestment recommendations.

The PBMA Panel

This panel should consist of members that are experts in the field of the area or programmes/ activities under scrutiny. The panel should also include relevant stakeholders, for example, those who deliver the programmes, patient and/or public representatives and partner organisations (charities and third sector organisations).

The PBMA exercise will require discussion, criteria setting and decision making; therefore, it may be useful to nominate a chairperson to facilitate these discussions, summarise them and assist the panel members to move through the stages of PBMA.

It may also be prudent to invite the manager or director who is endorsing the PBMA exercise to join the panel, so they can see the process in action and provide ‘buy in’ so other members see this as important activity and contribute to the exercise, so it is successful.

3.2 Stage 2 – Identify current activity and expenditure in those programmes

In order to identify current activity and expenditure a PBMA operational group is useful to undertake the tasks required to complete the stages of PBMA for example, gathering information on the activities identified.

Knowledge and expertise of the programmes under scrutiny would be beneficial. It is also advisable to have finance staff as members of this group so they can provide details of the proportional spend for each activity and calculate potential cost savings based on alternative scenarios of resource allocation.

Health economic expertise is also useful as part of the operational group, as health economists can guide the appraisal of economic evidence, guide the overall PBMA exercise and remind the panel of two key concepts when priority setting; ‘opportunity cost’ and ‘principle of the margin’.

Opportunity cost is the value of benefits foregone by not using resources in their next best alternative use.

Principle of the margin refers to altering the current pattern of expenditure to achieve the most efficient use of resources.

At the end of this stage there should be a list of activities that take place within the budget and information regarding the spend on each activity.

3.3 Stage 3 – Think of improvements

Once the programmes/ activities have been identified, the next step is to think of improvements to the current activity.

This could be done in a number of ways

- Liaising with stakeholders and experts to ask if there are things they would do differently.
- Seeking opinions of those who use the activities through; online surveys, engagement events, focus groups and suggestion/ comment boxes at activity sites.
- Exploring if previous audits or evaluations have been undertaken and see if any suggestions for improvements were made.
- Exploring published literature or national guidelines and standards.

Thinking of improvements could provide the panel with alternatives to current programmes, which could prove useful if the panel feel current activity is not achieving what the programme set out to do. This assists the PBMA panel when decision making and provides recommendations for future direction of travel. By using expert and stakeholder opinion there is an audit trail of why alternatives or improvements were suggested to the PBMA panel.

3.4 Stage 4 – Weigh up incremental costs and incremental benefits and prioritize a list

Before a list of candidate activities can be drawn up to present to the PBMA panel to discuss, the panel need to decide upon the criteria that will be used to appraise the activities. The criteria helps the panel to decide what is of importance such as the needs of the population, equity concerns, or value for money. The panel can discuss the purpose of the PBMA exercise and vote to reach a consensus. If the panel feel there is more than one purpose to the PBMA exercise, then ranking could be used to give a steer of the approach taken. This is useful when reporting the PBMA exercise as it provides the reader with the perspective of the PBMA.

The panel should also discuss how evidence regarding the activities will be appraised and prioritised. For example, is evidence of effectiveness and cost-effectiveness important, or is the reach of the activity important? Are they equally important or is some evidence more important than others? Once the panel have decided upon this evidence criteria, they need to rank it. The panel should rank the criteria or apply weightings, so as a group, the panel are in agreement of what is important when making their investment and disinvestment recommendations. This criteria can be revisited when appraising candidates and making investment and disinvestment recommendations.

The panel should also consider other criteria such as the time horizon that they expect return on the activities with this programme, pathway or budget. Are they looking for outcomes in the short, medium or long term?

In order to provide information of the incremental costs and benefits of each candidate that will be presented to the PBMA panel, the operational group will usually develop an evidence booklet.

Evidence Booklet

Evidence booklets should also contain information regarding the criteria set by the panel. For example if evidence of cost-effectiveness was a deemed a priority for appraisal by the panel, then available evidence should be included in the final evidence booklet. If there was no available evidence of the cost-effectiveness of a candidate, then this should be clearly stated in the evidence booklet.

There needs to be a consistent approach when gathering evidence to present to the PBMA panel to allow cross-comparisons. This is important as activities can span ages, sectors, and geographical areas. There also may be a range of activities taking place to tackle a particular goal.

Evidence can be gathered through systematic reviewing to identify published evidence of effectiveness or cost effectiveness that underpin an activity, case studies, local evaluations including forecasting or modelling by finance teams and focus groups. It may also be helpful to liaise with staff who deliver the activities to explore wider outcomes such as tackling inequalities.

The evidence booklet should also describe the quality of the evidence presented.

Quality of evidence

The quality of evidence is also important to report as part of the evidence gathering so the PBMA panel are clear on the limitations of evidence presented. This is especially important as it is sometimes difficult to find evidence for specific programmes that may have been developed locally or are a hybrid of two or more evidence based programmes.

There are critical appraisal tools such as those published by **The Joanna Briggs Institute** and **The Critical Appraisals Skills Programme** (CASP), and subject specific quality appraisal checklists (e.g. **checklist for a sound economic evaluation**), which can be used to critically appraise the evidence found.

In order to make the evidence easy to interpret by the PBMA panel, the operational group may wish to grade or rate the evidence using a traffic light system, a numerical system or another approach. Below is an example of how a traffic light system could be used for published evidence and the degree of equality/ equity focus.

Example of traffic light grading for published evidence

Green	Amber	Red	Not Applicable
Published evidence of good quality as appraised by...	Published evidence of fair quality as appraised by…	Published evidence of poor quality as appraised by…	No published evidence found.

Example of traffic light grading for the degree of equality/equity focus

Green	Amber	Red	Not Applicable
Activity is specifically designed to reduce health inequalities or states within its aims to reduce inequalities, and there is evidence of this.	Activity states it aims to reduce inequalities but there is no evidence of this.	Activity has no stated impact on or aims that are likely to reduce inequalities.	Activity is not expected to impact on inequalities.

The PBMA panel may have a number of candidates to discuss; therefore, it is important to keep the evidence booklet as succinct as possible. A wider methodology booklet can be developed to explain in further detail how the evidence was gathered, appraised and summarised. It is advisable to share both the evidence booklet and the accompanying methodology with the PBMA panel, at least two weeks before the panel are due to discuss candidates and make recommendations on resource reallocation.

For example, methodology booklets and evidence pro-formas.

Example evidence booklets/ pro-formas.

Title of programme, activity or area of spend	
Description	
Life course stage or Age of recipients	
Spend	
Evidence of effectiveness	
Evidence of cost-effectiveness	
Evidence of previous evaluations	
Potential risks if changes are made	
Likely impacts if changes are made	
Links to policy	
Additional notes (e.g. equity considerations, inequality, comments from service users)	

3.5 Stage 5 – Consult widely

By this stage the evidence will have been gathered and administered to the PBMA panel. It is recommended that the PBMA panel have plenty of opportunity and time to discuss the budget, activities and review the related evidence.

Providing information ahead of time allows the panel to digest the information so they are prepared for discussions. If possible face-to-face discussions are ideal, but virtual meetings could be used where this is not possible. It is also advisable that finance staff and members of the operational group attend the discussion meetings as they can answer questions that arise regarding information presented to the panel.

3.6 Stage 6 – Decide on changes

The PBMA panel will be asked to discuss each candidate in turn, keeping in mind their agreed aim(s) of the PBMA exercise and ranked evidence criteria.

The panel will also need to decide on their voting criteria, are they making dichotomous “invest” or “disinvest” recommendations or include options to “partially invest” or “partially disinvest”. Would the panel prefer to rank priorities and decide to disinvest in low priorities and invest in high priorities?

How the panel vote and decide upon consensus will also need to be agreed ahead of voting. The panel could use voting techniques, conduct a ranking exercise or ask the chairperson to gauge the consensus based on the discussion.

When using voting as a technique, the panel may choose to conduct an open vote where each member is aware of co-members' recommendation or choose to anonymously vote. Anonymous voting is championed in PBMA as it allows panel members to make independent recommendations on each candidate.

The panel may like to have the results presented immediately after each vote, so the panel can see whether or not consensus was reached. If a consensus has not been reached or votes are spread across the options, the panel could re-discuss the candidate and choose to re-vote, if appropriate.

Once the panel have agreed which candidates will be disinvested in and which will be invested in, they need to state the amount of money they wish to disinvest or invest**.**

3.7 Stage 7 – Effect the changes

Once the PBMA panel has voted and agreed their recommendations for investment and disinvestment, these recommendations need to be communicated with their directors/ managers who can action them.

The panel may wish to present the recommendations verbally or in a report. However, the panel should keep in mind to present the context of the discussions, especially if the director/ manager was not able to participate in the panel. The reasons why a particular aim was agreed or a re-vote required will need to be reported, so decision making required throughout the PBMA exercise to arrive at the recommendations is transparent.

3.8 Stage 8 – Evaluate progress

Once the recommendations from the current PBMA exercise have implemented, see the framework as a useful tool to audit progress of the programme/ budget in the future.

It is likely budgets will change, the population the activities serve will change, new initiatives/ programmes will become available and pathways may change. These changes could highlight activities that could be considered a better use of resources than some of the candidates presented in the previous PBMA exercise. Individuals should be guided by these changes to determine the opportune moment to conduct another PBMA exercise.

Example meeting strategy for PBMA

From previous experience of conducting PBMA momentum is key to ensure the framework is completed successfully. Ideally around 6-8 months to complete an exercise would allow sufficient time for all stages to be conducted.

The running order presented should provide a steady pace for the PBMA panel to work through the PBMA stages 5-7.

- The initial face-to-face or virtual meeting should outline the need for conducting a PBMA (i.e. state the problem) and explain the PBMA framework and process.

- The next face-to-face or virtual meeting should describe the budget in detail, and set the boundaries of the PBMA exercise. It is essential finance staff lead this meeting.

- The next face-to-face or virtual meeting should discuss and set the aim of the PBMA exercise. This meeting should also discuss and rank/ weight evidence criteria so the PBMA panel and chairperson are clear about what matters the most to the panel when appraising and discussing candidates for investment/ disinvestment recommendations.

- The next face-to-face or virtual meeting should discuss and make recommendations on the candidates for investment/ disinvestment. This meeting could be split into two meetings if there is a large number of candidates to discuss. At the end of this meeting the panel should be clear on which candidates they wish to invest/ disinvest in, how much money they wish to invest/ disinvest and whether they wish to invest any released monies into current activity or seek alternatives.

- The next face-to-face or virtual meeting should re-visit the recommendations of the previous meeting and agree the recommendations that will be presented to the decision maker(s) following the PBMA exercise. It should also be agreed how the recommendations will be presented (e.g. report, verbal presentation etc.)

4. Examples of published PBMA exercises

Micro health settings examples:

Henderson LR, Scott A. **The costs of caring for stroke patients in a GP-led community hospital: an application of programme budgeting and marginal analysis**. Health Soc Care Community 2001;9 (4):244–54.

Mitton C, Donaldson C, Shellian B et al. **Priority setting in a Canadian surgical department: a case study using program budgeting and marginal analysis**. Can J Surg 2003;46 (1):23–9.

Bolaji S. **Programme budgeting marginal analysis (PBMA) for COPD PHCN Casebook**.

Charles JM, Brown G, Thomas K, et al. **Use of programme budgeting and marginal analysis as a framework for resource reallocation in respiratory care in North Wales, UK**. Journal of Public Health (Oxford, England). 2016 Sep;38(3):e352-e361. DOI: 10.1093/pubmed/fdv128.

Macro health settings examples:

Haas M, Viney R, Kristensen E, Pain C, Foulds K. **Using programme budgeting and marginal analysis to assist population based strategic planning for coronary heart disease**. Health Policy. 2001 Mar;55(3):173-86. doi: 10.1016/s0168-8510(00)00119-6.

Edwards RT, Charles JM, Thomas S et al. **A National Programme Budgeting and Marginal Analysis (PBMA) of Health Improvement Spending Across Wales: Disinvestment and Reinvestment Across the Life Course**. BMC Public Health 2014;14:837.

Holmes RD, Steele JG, Exley C, Vernazza CR, Donaldson C. **Use of programme budgeting and marginal analysis to set priorities for local NHS dental services: learning from the north east of England**. J Public Health (Oxf). 2018 Dec 1;40(4):e578-e585. doi: 10.1093/pubmed/fdy075. PMID: 29726998.

5. Contact information

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